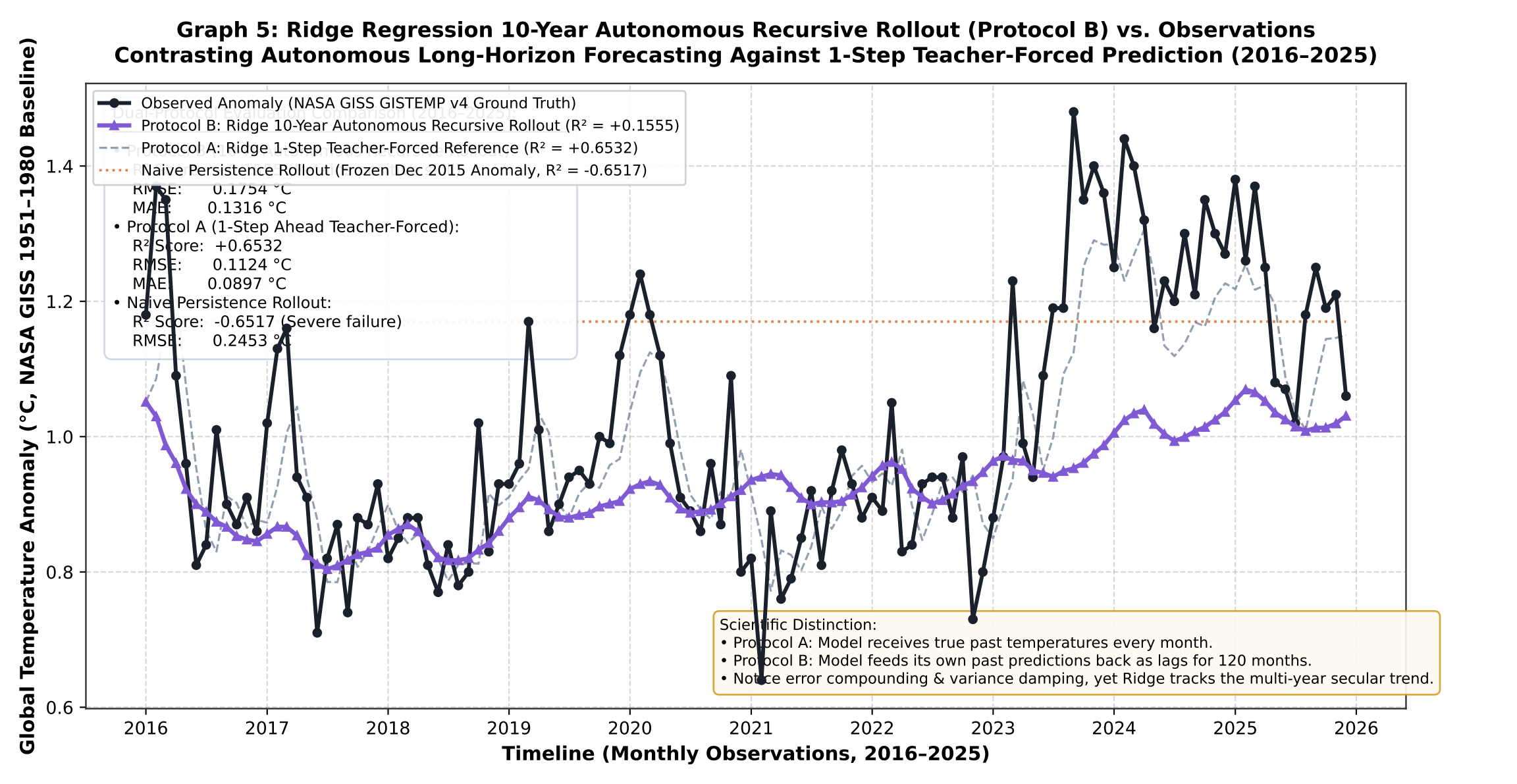
Graph 5: Ridge Regression 10-Year Autonomous Recursive Rollout (Protocol B) vs. Observations
Contrasting Autonomous Long-Horizon Forecasting Against 1-Step Teacher-Forced Prediction (2016–2025)
Observed Anomaly (NASA GISS GISTEMP v4 Ground Truth)
Protocol B: Ridge 10-Year Autonomous Recursive Rollout (R² = +0.1555)
Protocol A: Ridge 1-Step Teacher-Forced Reference (R² = +0.6532)
Naive Persistence Rollout (Frozen Dec 2015 Anomaly, R² = -0.6517)
RMSE: 0.1754 °C
MAE: 0.1316 °C
• Protocol A (1-Step Ahead Teacher-Forced):
R² Score: +0.6532
RMSE: 0.1124 °C
MAE: 0.0897 °C
• Naive Persistence Rollout:
R² Score: -0.6517 (Severe failure)
RMSE: 0.2453 °C
Scientific Distinction:
• Protocol A: Model receives true past temperatures every month.
• Protocol B: Model feeds its own past predictions back as lags for 120 months.
• Notice error compounding & variance damping, yet Ridge tracks the multi-year secular trend.
Global Temperature Anomaly (°C, NASA GISS 1951–1980 Baseline)
Timeline (Monthly Observations, 2016–2025)
0.6
0.8
1.0
1.2
1.4
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026